

| Assessment | Test | Domain        | Skill                  | Difficulty                               |
|------------|------|---------------|------------------------|------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | Hard <div></div> <div></div> <div></div> |

Question

$$\frac{x^2 - c}{x - b}$$

In the expression above, b and c are positive integers. If the expression is equivalent to  $x + b$  and  $x \neq b$ , which of the following could be the value of c ?

Answer

- A. 4
- B. 6
- C. 8
- D. 10

Correct Answer: A

Rationale

Choice A is correct. If the given expression is equivalent to  $x + b$ , then  $\frac{x^2 - c}{x - b} = x + b$ , where x isn't equal to b. Multiplying both sides of this equation by  $x - b$  yields  $x^2 - c = (x + b)(x - b)$ . Since the right-hand side of this equation is in factored form for the difference of squares, the value of c must be a perfect square. Only choice A gives a perfect square for the value of c.

Choices B, C, and D are incorrect. None of these values of c produces a difference of squares. For example, when 6 is substituted for c in the given expression, the result is  $\frac{x^2 - 6}{x - b}$ . The expression  $x^2 - 6$  can't be factored with integer values, and therefore  $\frac{x^2 - 6}{x - b}$  isn't equivalent to  $x + b$ .